Change Log: v2 → v3

Gallup Engagement Agent · app (Selvi's second version) + synthetic data tools · October 2026

> Seven issues were found when the v2 test sets were run against the app. All seven are fixed. The 5K, 2M, null, seed and dirty data files are byte-identical to v2; only the weak test set was regenerated. Health reports were regenerated with health check 3.0 (same 45 checks, same results).

Fixes

#	Issue	Impact	Fix	Files	Evidence
1	**Health gate leak: a question that named no outcome skipped the gate**	HR could get an interpretation of data the health check had blocked (dirty_C: 3 of 5 questions reached the LLM)	New `health_gate()`: every outcome blocked → refuse any question and list the failed checks; asked outcome blocked → refuse it and name the outcomes still available; no outcome named and some blocked → answer from allowed outcomes only, evidence carries an exclusion note (prompt rule 42). `outcome_is_allowed(analysis, None)` no longer returns True when everything is blocked.	`src/agent.py`	v2: on dirty_C, 3 of 5 test questions reached the LLM. v3: dirty_A, B and C × 5 questions, all 15 refused, 0 LLM calls. Unit tests: 4.
2	**Models ran before the health check**	dirty_A crashed with KeyError 'performance_rating'; blocked outcomes still produced findings	Health report is read first. All outcomes blocked → no model is fitted and an empty, BLOCKED analysis is returned. Some blocked → only allowed outcomes are fitted, so blocked findings never reach payloads, dashboards or the LLM. Outcome columns that wrongly sit in the survey file are dropped before the merge.	`src/tools/analytics.py`	dirty_A/B/C: status BLOCKED, no crash. Planted date error in performance only: status PARTIAL, turnover and absence still analysed. Unit tests: 2.
3	**Uploads used a placeholder "all pass" health report**	Uploaded files were never checked: errors not blocked, X7 nonresponse caveat missing, small teams not flagged	`health_check.py` refactored into functions (`run_checks`, `check_dataframe`, `check_folder`). New `analyze_upload(df)` checks the whole upload (every wave, respondents and non-respondents), then models the model wave. Used by the Ask AI, Recommendations and analysis-runner pages; the development dataset uses its own report. UI shows a red/orange badge when outcomes are blocked.	`health_check.py`, `analytics.py`, `agent.py`, `ui.py`, `app_pages/agent.py`, `app_pages/recommendations.py`	Refactored check gives identical results to v2 on all 8 datasets (45 checks each). 5K uploaded as one table: NR_LEAVERS and COV_SMALL_TEAMS now WARN. Upload with out-of-range scores: BLOCKED. Unit tests: 3.
4	**X4 threshold not reported**	HR not told that absence jumps when Q01 is below 3; one trap point lost	`threshold_diagnostics()` reports the Q01-below-3 term of the absence model (Matrix A) when p < 0.05 and outside the no-effect band. Attached to the matching payload as `related_patterns` and to the agent evidence (prompt rule 43).	`analytics.py`, `agent.py`	Rate ratio 1.805 on 5K and 1.832 on 2M; answer key realized values 1.805 and 1.832.
5	**X5 new-hire subgroup not reported**	HR not told recognition matters more for new hires; another trap point lost (max 6/8)	`subgroup_diagnostics()`: one turnover model with all 12 item × new-hire (tenure < 2 years) interactions, FDR-corrected; reports odds ratios for new hires and others separately.	`analytics.py`, `agent.py`	Only Q04 flagged. 5K: OR 0.52 new hires vs 0.83 others (log-OR difference −0.46; key −0.44). 2M: 0.50 vs 0.71 (−0.36; key −0.36).
6	**No API retry**	A single HTTP 429 or empty answer stopped a validation run	`call_with_retry()`: retries 429, 408/409/5xx, timeouts, connection errors and empty answers up to 3 times, waiting 2, 4, 8 s or the server's Retry-After. Authentication and other client errors are not retried.	`src/agent.py`	Unit tests: 4 (two 429s then success; empty then success; gives up after 3 retries; no retry on 401).
7	**Weak test set: T5 inside the no-effect band (our data-side error)**	T5 realized at IRR 0.94, so a correct agent does not report it while the answer key calls it a true effect	Generator 2.1: weak T5 planted at IRR 0.87. Validator: new check that every true effect's realized value lies outside the band (T1–T5).	`generate_synthetic_data.py`, `validate_synthetic_data.py`	Weak T5 realized 0.886; the app now reports Q05 → absenteeism as ROBUST (0.886). Weak 36/36; all other sets unchanged and pass.

Open items (not changed in v3)

Item	Detail
Pre-existing test failure	`tests/test_recommendations.py::test_team_plan_is_plain_language[position=-1]` fails with KeyError nan in v2 as well (not touched).
Analysis wave	The app models the newest wave with outcomes (wave 2 for v2 data); the answer key and `analyze_dataset` use wave 1. Decide which wave is graded.
Dashboard sections under a partial block	Overview, financials and trends are computed from the raw table. When one outcome is blocked, the agent is told not to discuss it (prompt rule 42), but the pages still show it. A page-level filter is a follow-up.
Not verifiable here	Streamlit, altair and openai are not installed in the build environment. Logic was tested with stand-ins: 180 of 181 tests pass (the failure above). Run `python -m pytest -v` and a browser smoke test on a laptop before merging.

Tests

New file `tests/test_v3_fixes.py` (16 tests, one or more per fix). Whole suite: 180 of 181 pass in the build environment. Data validator: 5K 36/36, 2M 36/36, weak 36/36, seed2 36/36, seed3 36/36, null 19/19, dirty_A 10/10, dirty_B 10/10, dirty_C 5/5.